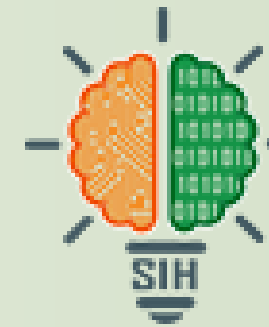


GRP_4_SMTH

SMART INDIA HACKATHON 2025



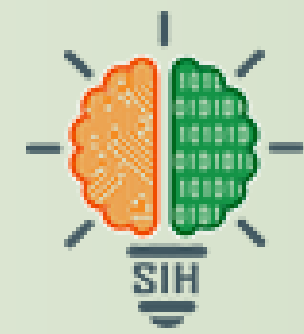
SMART INDIA
HACKATHON
2025

- Problem Statement ID – **SIH25015**
- Problem Statement Title- **Intelligent Pesticide Sprinkling System Determined by the Infection Level of a Plant**
- Theme -**Agriculture, FoodTech & Rural Development**
- PS Category- **Hardware**
- Team Name (Registered on portal) : **GRP_4_SMTH**
- Team Id - **56016**



GRP_4_SMTH

CROPGRID AI: GRID-BASED INFECTION MAPPING & PRECISION SPRAYING SYSTEM



SMART INDIA
HACKATHON
2025

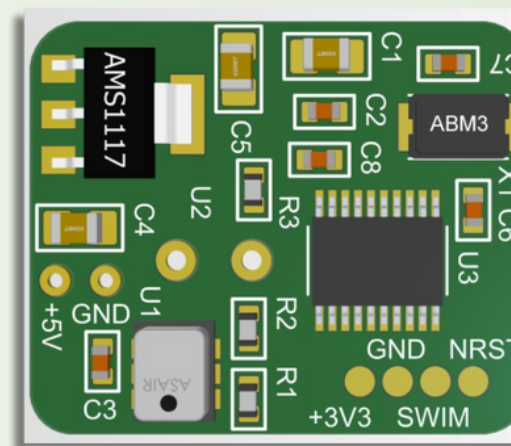
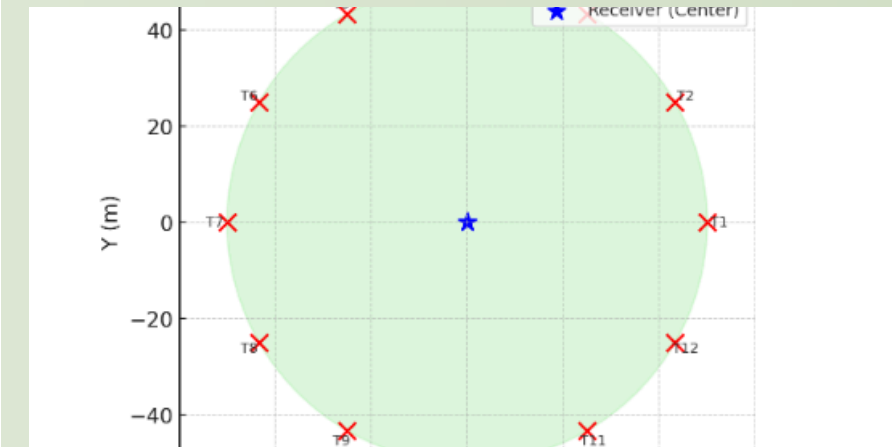
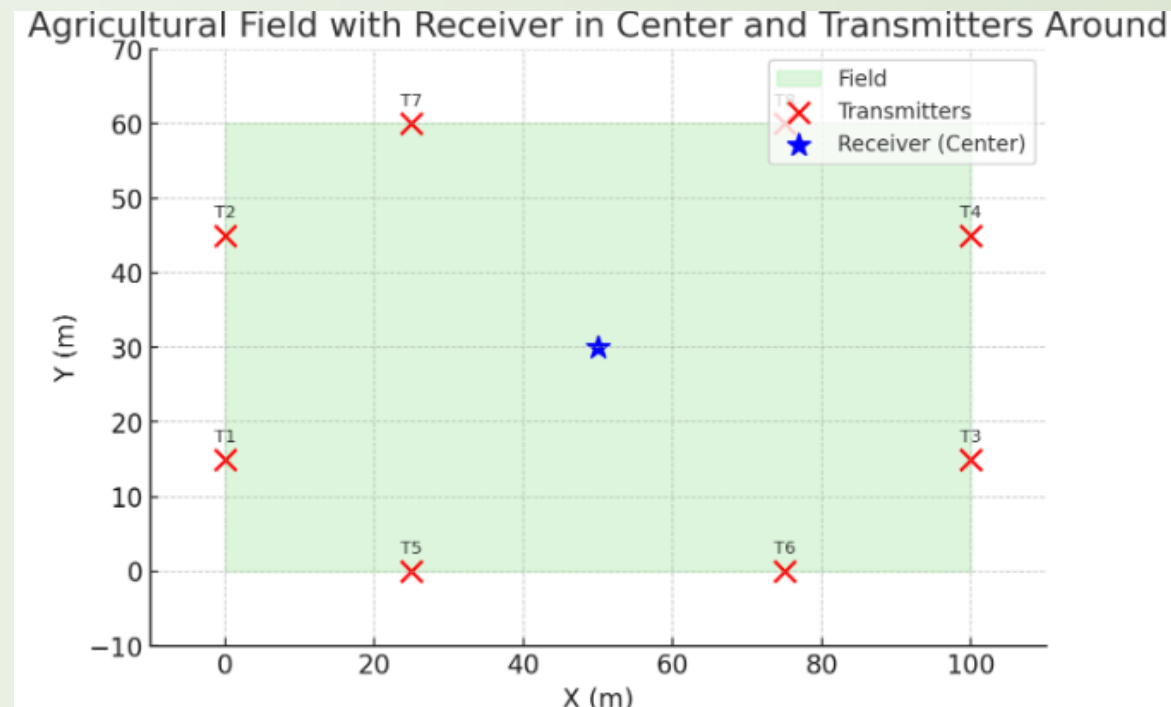


Combination of optical and contact Imaging will help in determining the overall health of field crops

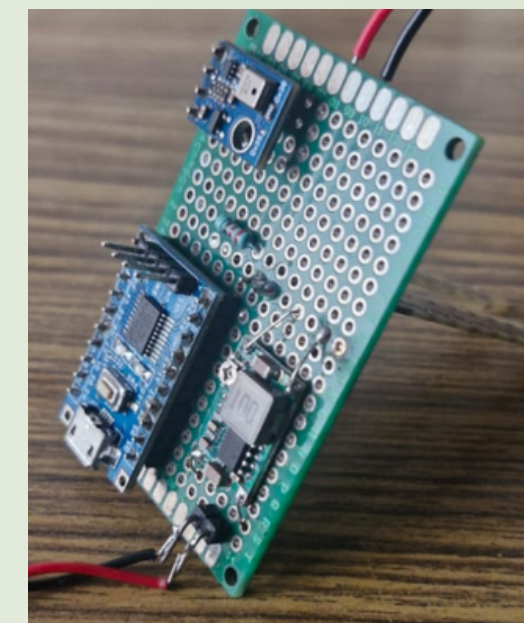
We've developed an integrated **AI + IoT precision agriculture platform** that transforms traditional farming through intelligent automation.

Our system uses **active NDVI sensors** with controlled light transmitters and receivers to measure plant reflectance, detecting stress and pest infestations even during nighttime operations.

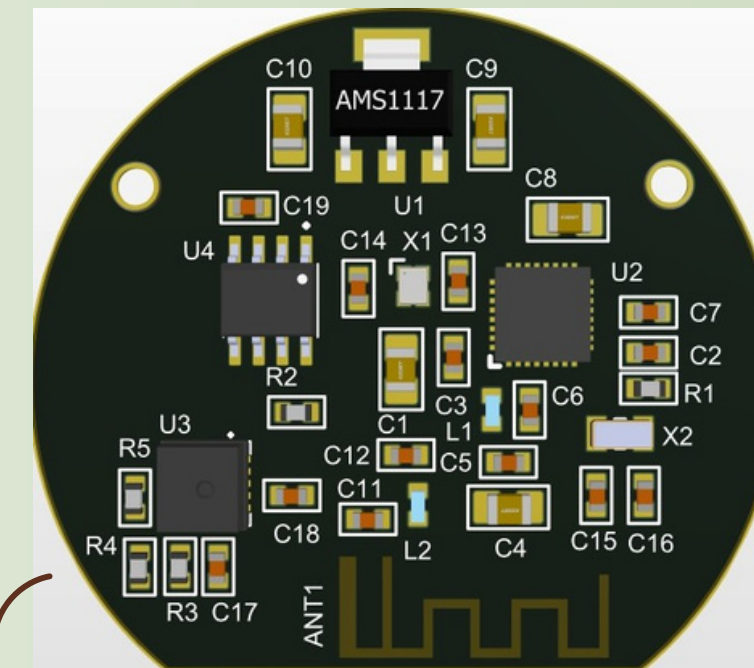
Real-time data flows to our cloud-based machine learning engine, which analyzes crop health patterns and pinpoints affected areas with surgical precision.



GND MODULE



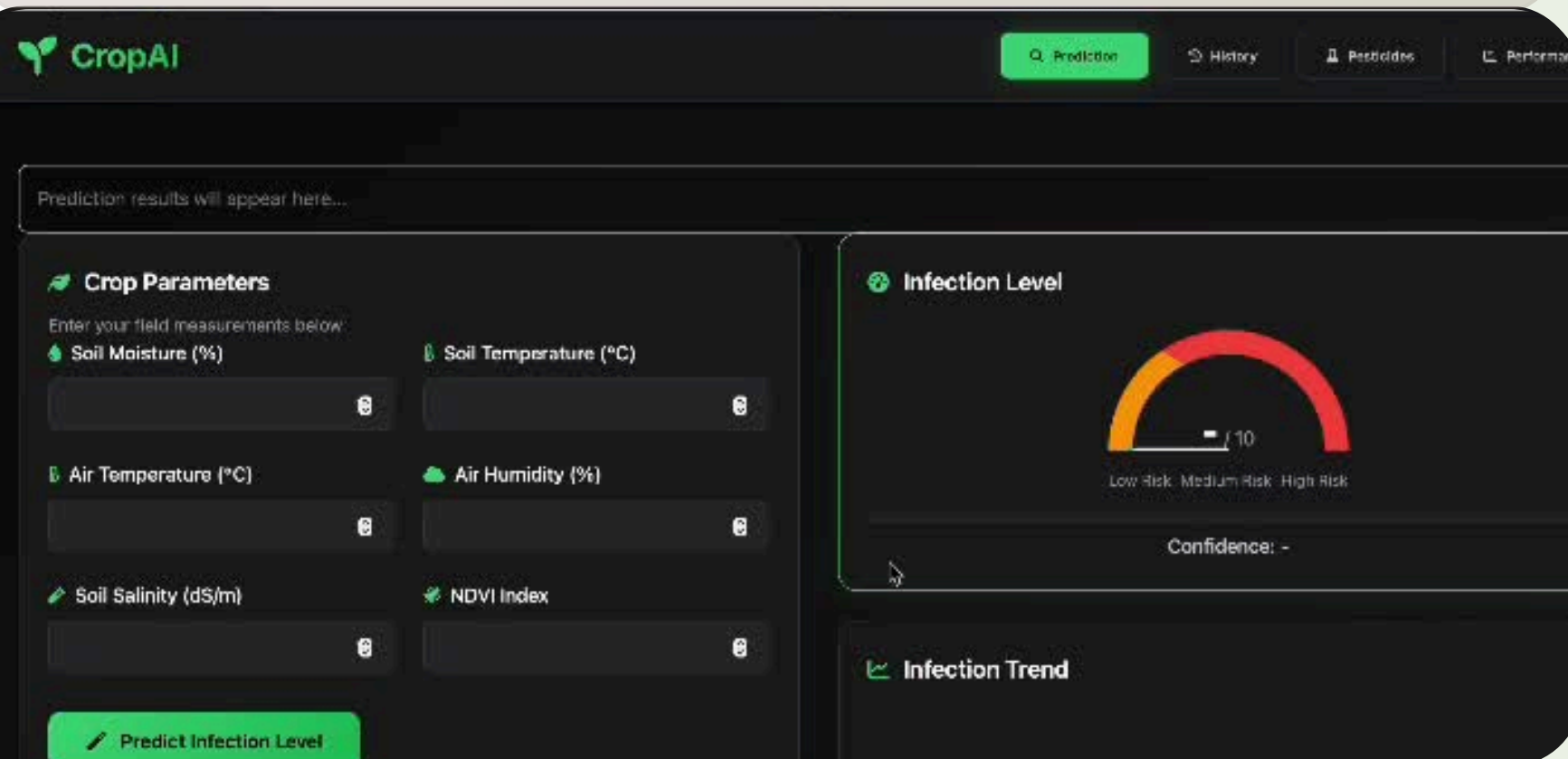
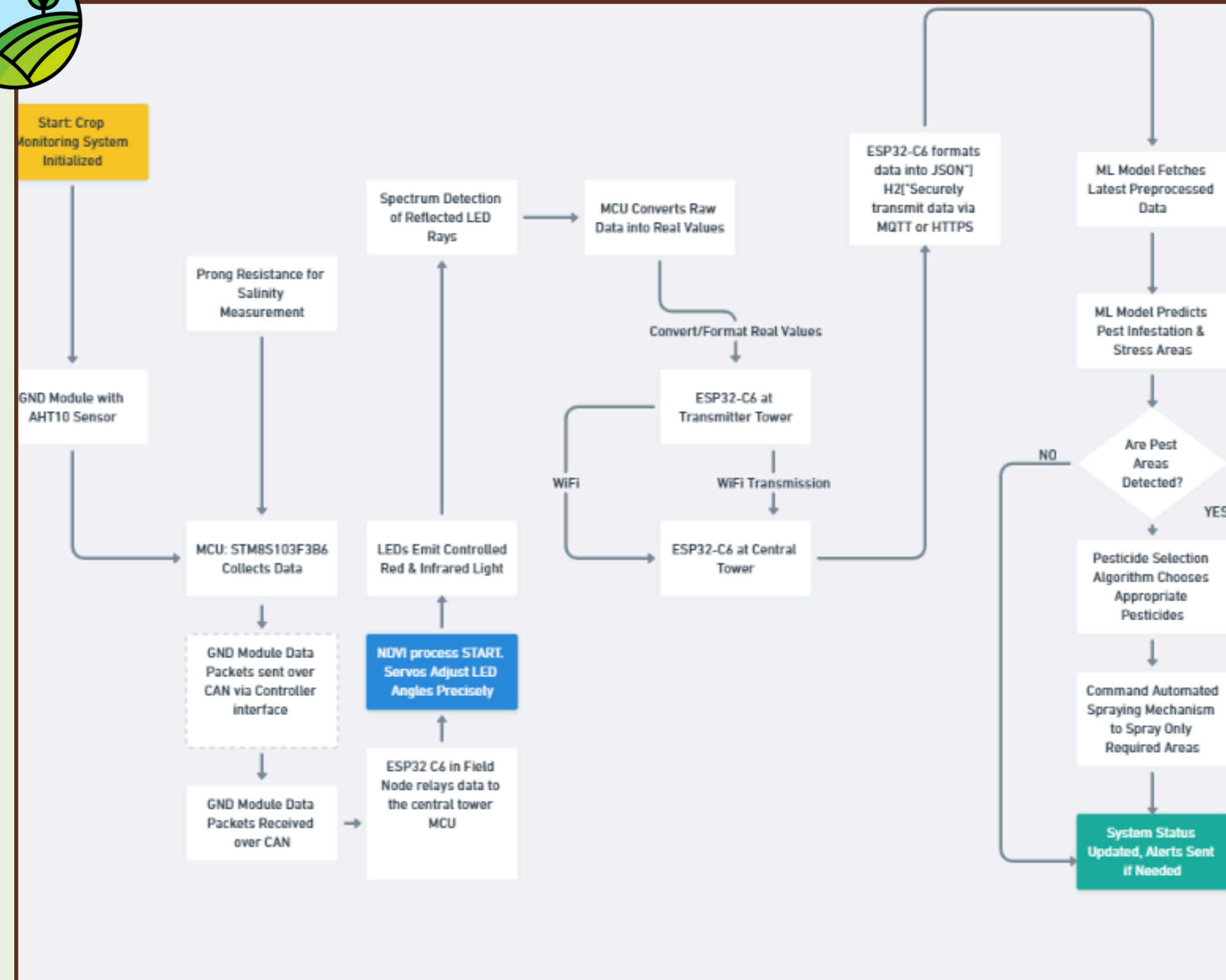
PROTOTYPE



RECEIVER

EMBEDDED SYSTEM

- **GND Modules (STM8 + sensors):** Soil moisture, temp, salinity.
- **LED Arrays + Servos:** Emit RED/IR for NDVI sensing.
- **AS7263 Spectrum IC:** Captures reflectance spectrum.
- **ESP32 MCU(Field Nodes and Central Towers):** Collects data, sends via ESP-NOW.
- **Cloud ML + Sprayers:** Analyze → decide → targeted spray.
- **AC to DC Conversion:** Hi Liink + filter cap.
- **Buck Converter:** Step-down DC (e.g., 12V → 5V/3.3V).
- **STM8 MCU (Ground Modules)** collect soil and crop data.
- **An XGBoost model** analyzes this data to predict infection risk levels.
- Based on the risk, the system recommends targeted pesticides (Neem → Sulfur Dust → Copper/Mancozeb) and displays results on a web platform.





ECONOMIC

- **Affordable with Support:** ₹30-40k viable via KCC loans (7% interest) + SMAM 40-50% subsidy = ₹15-20k net cost
- **Strong ROI:** 30-50% pesticide savings + premium crop pricing = payback within 2-3 seasons
- **Government Funding:** Multiple sources - AIF (3%), KCC (₹3 lakh), PM-KUSUM (75% solar subsidy)



TECHNICAL

- **Proven Tech:** Active NDVI sensors 97.8% accurate, day/night operation, successful IoT integration.
- **Ready Infrastructure:** Cloud/edge computing available, ML algorithms proven, automated sprayers exist.



MARKET

- **Huge Market:** 86% farmers are small/marginal, ₹1.67 lakh crore annual fertilizer subsidy shows urgent need.
- **Multi-stakeholder Win:** Farmers save money, consumers get safer food, government reduces subsidy burden.



IMPLEMENTATION CHALLENGES AND SOLUTIONS:

- **Connectivity** → On-device processing reduces internet dependency for real-time decisions
- **Training** → Extension Partnership: Use existing Krishi Vigyan Kendras and PAU for farmer education
- **Maintenance** → Local Network: Train technicians via agricultural universities, mobile service units



FOR FARMERS

- **Cost Reduction:** Significant savings on pesticide expenses through precise, need-based application rather than blanket spraying
- **Extended Chemical Life:** Pesticides last longer due to efficient usage, reducing frequent purchasing costs
- **Higher Profit Margins:** Premium pricing for residue-free, healthier crops in quality-conscious markets
- **Long-term ROI:** Better return on investment through reduced input costs and improved crop value
- **Sustainable Farming:** Enhanced soil and water quality preservation for long-term farm productivity



FOR CONSUMERS

- **Healthier Food:** Access to crops with minimal pesticide residues, reducing health risks from chemical contamination
- **Premium Quality Produce:** Higher nutritional value and better taste in crops grown with precision agriculture
- **Food Safety Assurance:** Reduced exposure to harmful chemicals through targeted pesticide application
- **Environmental Benefits:** Contribution to cleaner ecosystems and reduced environmental pollution
- **Market Value:** Access to certified organic or low-residue produce that commands premium prices in health-conscious markets

- "NDVI technology can detect crop stress up to 10 days before visible symptoms appear, enabling faster intervention" - Farmonaut Research (2025)
- "Higher temperature (along with high humidity) is linked to enhanced disease severity of plant pathogens" - Singh et al., Nature (2023)
- "Crops grown on saline soils suffer high osmotic stress, nutritional disorders and reduced crop yield" - Shrivastava & Kumar, PMC (2014)
- "IoT for Real-Time Pest Detection in Agriculture" Prachi Nalawade(2025)
- "Traditional methods of crop health assessment, such as manual inspections and laboratory analyses, are often labor-intensive, time-consuming, and prone to human error" - Dhingra et al., International Archives of Photogrammetry (2025)
- "Elevated temperature stress leads to desiccation and makes the plant prone to parasitic attack" - Pathak et al
- Datasheets of ICs referred for indigenous development of modules from ST , MICROCHIPS.
- "NDVI changes as early warning for disease, pest outbreaks":
<https://farmonaut.com/remote-sensing/ndvi-guide-boost-crop-health-with-vegetation-index>
- Characterizing vegetation indices derived from active and passive sensors:
<https://www.researchgate.net/publication/233429767> Characterizing vegetation indices derived from active and passive sensors

